

BUILDING STOCK ENERGY WORKBENCH

Valencia Simulation Report

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Run: ALL_VALENC-A_REAL

Document status: Provisional method edition; the run is active. Final values, figures and selected-building evidence will be inserted only after the process ends and the aggregate settles.

This report documents the annual Valencia building-stock simulation executed in Building Stock Energy Workbench (BSEW). It is designed to answer four questions: what evidence entered the run, which modelling and execution parameters were applied, which quality gates controlled acceptance, and what the settled results support academically.

PENDING FINAL EVIDENCE — DO NOT CITE. No running total in the active ledger is reproduced in this edition. The publication status will change only after running: false, process absence, final aggregate settlement and cross-file reconciliation have all been verified.

Suggested citation

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1. Executive interpretation

ALL_VALENC-A-REAL is a full-scope annual simulation of the supplied Valencia building stock under one recorded verified model profile. BSEW converts cadastral and dwelling evidence into simplified OpenStudio models, executes EnergyPlus, records terminal evidence per building and derives aggregate energy, operational-carbon, coverage, geometry-quality and QA summaries.

The report will publish final values only when all latest ledger records reconcile with the settled scope. The principal result classes will be total site energy, whole-site energy intensity under explicitly named residential-area denominators, end-use energy, operational carbon, cluster/district summaries, coverage, and diagnostic status.

The results are simulations under recorded assumptions. They are not measured utility consumption, per-building calibration, an energy certificate, regulatory compliance, embodied carbon or a guaranteed forecast of future operation. The verified profile establishes method consistency against the project's reference evidence; it does not establish measurement validation for every stock building.

1.1 Publication decision

The final report will assign one stock-level publication decision after reconciling successful, failed, QA-rejected and excluded records. Building-level records remain individually classifiable as PASS, ACCEPTED WITH CLASSIFIED DIAGNOSTICS, FAILED / DO NOT USE, NOT ASSESSED or INCOMPLETE EVIDENCE – DO NOT CITE.

1.2 Final evidence gate

Table 1. Final evidence gate

Gate	Required final evidence	Provisional state
Process	Run reports false and recorded PID is absent	Pending
Scope	Final terminal records reconcile to recorded scope	Pending
Aggregate	Final aggregate exists, is readable and is not partial	Pending
Identity	Latest accepted rows share the intended run/profile/input identity	Pending
Exports	CSV, GIS, heat map and manifest reconcile to the ledger	Pending
Review	QA, diagnostics, limitations and selected buildings reviewed	Pending

2. Research objective and analytical boundary

The simulation provides reproducible evidence for the spatial and typological distribution of modelled final site energy across the supplied Valencia stock. It supports comparison among recorded building clusters and declared area bases, inspection of preparation and EnergyPlus evidence at building level, and production of traceable row-level and spatial exports.

The analytical unit is the building or cadastral reference represented by the latest terminal ledger record. Aggregate values include only records admitted by the stated result and QA contract. Excluded and failed records remain visible in coverage evidence and are not interpreted as zero energy.

2.1 Questions the report can answer

- What final site energy and operational emissions were simulated for the represented stock?
- How do results distribute across building clusters and, where supported, districts?
- How do geometric and cadastral residential-area denominators change intensity interpretation?
- Which buildings failed or were excluded, and what bias can or cannot be quantified?
- Which preparation, geometry, occupancy and QA assumptions constrain interpretation?

2.2 Questions it cannot answer alone

- Whether a specific building matches its utility bills or actual occupancy.
- Whether the stock is calibrated against measured consumption.
- Whether a simulated difference is causally attributable to an intervention.
- Whether operational factors represent embodied or whole-life carbon.
- Whether simplified one-zone-per-storey geometry captures all local thermal diversity.

3. Run identity and execution record

The final report will read identity from the settled run directory and authoritative API response, not from the currently active Files page. The run name is a human-recognisable label; cryptographic fingerprints establish the resolved scientific identity.

Table 2. Run identity and execution record

Identity element	Authoritative source	Final value
Run name	Run directory and API route	ALL_VALENC-A_REAL
Run identity	run_config.json / ledger rows	Pending final reconciliation
Verified profile	run_config.json / ledger rows	Pending final reconciliation
Climate fingerprint	Run configuration and ledger	Pending final reconciliation
Template fingerprint	Run configuration and ledger	Pending final reconciliation
Policy fingerprint	Run configuration and ledger	Pending final reconciliation
Stock-source fingerprint	Run configuration and ledger	Pending final reconciliation
Runner schema	Run configuration and ledger	Pending final reconciliation
Execution segments	Process records, logs and run configuration	Pending final reconciliation of start/resume worker counts and retention

The current production interface submits six workers and full per-building retention for a new UI request; these are not user-editable controls on the Run page. The Valencia run has an interruption/resume history, so this report will not assume that every execution segment used the UI default. The final execution table will disclose the worker count and retention recorded for each recoverable segment, including any changed resume count. Worker count affects throughput rather than the frozen model inputs, but it remains part of execution provenance. Wall-clock duration, accumulated worker CPU time and median/mean/max building duration will be reported only from settled process evidence.

3.1 Period

The Valencia run is annual. Final energy is reported as kWh/year or GWh/year; intensity as kWh/m²·year; operational carbon as kgCO₂/m²·year or tCO₂/year where the corresponding denominator is explicit. No microclimate event values will be mixed into this annual ledger.

4. Authoritative inputs

The run resolves its active input set once and records content identities. Absolute workstation paths are excluded from this report. Logical dataset role, preserved filename, validation contract and SHA-256 evidence are retained.

4.1 Valencia cadastre

The building GIS supplies parcel/building geometry and attributes used for identity, scope, storeys, population, dwellings, cluster, district/context and footprint preparation. It is an administrative/research source, not a geometric survey. The final table will record managed filename, feature count, CRS, geometry type, snapshot hash and field-role policy.

4.2 Tipo15 dwelling ledger

The companion semicolon-delimited Latin-1 CSV supplies 31_pc, 252_planta, 428_uso and 442_sup_Residencial. It supports parcel join, floor/use interpretation and the cadastral residential-area denominator. Multiple dwelling records per parcel are expected. Missing joins and fallback use will be quantified rather than converted to zero.

4.3 EPW and DDY climate pair

The annual EPW supplies hourly weather and site metadata; the matched DDY supplies winter and summer design days. The pair is validated and activated atomically. Slab-contact ground temperature and water-mains temperature are recorded separately because an EPW does not authoritatively provide both model inputs.

4.4 OpenStudio template

The OSM template provides role-bound constructions, schedules, thermostats, space types and required system objects. Its fingerprint identifies the source library used to build every accepted model; it is not a measured representation of each building.

Table 3. OpenStudio template

Dataset role	Preserved filename	Contract	SHA-256 / fingerprint
Building GIS	Pending final run evidence	GIS snapshot + active policy	Pending
Tipo15	Pending final run evidence	tipo15-v1	Pending
Annual EPW	Pending final run evidence	epw-annual-v1	Pending
DDY	Pending final run evidence	ddy-design-days-v1	Pending
OpenStudio template	Pending final run evidence	template-roles-v1	Pending

5. Software and verified profile

The documented toolchain is Python 3.13, OpenStudio 3.11.0 and EnergyPlus 25.2.0 [1]–[3]. The final report will record the exact patch versions and software commit from the release and run evidence. Dependency presence is a reproducibility condition, not evidence of physical accuracy.

The verified profile pins the model-building sources, template identity, policy and acceptance evidence. Profile verification means that the modelling method passed the project's fixed reference gates. It does not mean that Valencia's individual buildings were calibrated to measurements.

5.1 Profile-integrity rule

Every accepted ledger row must carry the intended profile fingerprint. Source drift during execution is not silently accepted. A mechanically combined ledger containing more than one profile cannot be published as one homogeneous run without explicit separation and warning.

5.2 Frozen and derived evidence

Physical model outputs are produced by the pinned simulation chain. Reporting may format units and labels, select latest records, create deterministic aggregates and check duplicate evidence for conflicts. It must not silently recompute historical physical results with today's changed model rules.

6. Building preparation and geometry

The pipeline transforms supplied polygons into tractable building masses, identifies context and party surfaces, applies storey rules, assigns ground use, creates openings where exterior walls exist, and records fidelity evidence.

6.1 Footprint preparation

Simplification fidelity is measured as symmetric difference divided by raw area. The configured bound and selected tolerance are recorded per building. A refined tolerance or as-drawn fallback protects geometry when the default tolerance would exceed the fidelity bound. A high fidelity score means closeness to the supplied polygon under this transformation; it does not prove that the source polygon is survey-accurate.

6.2 Storeys and top-storey allocation

The report distinguishes total levels, residential levels above ground, effective residential storeys, built storeys and top-storey fraction. A recorded storey snap may remove a resolution-scale sliver near an integer boundary. Dwelling loads on a partial top storey are scaled according to the recorded fraction; the physical model still uses whole geometric storeys.

6.3 Zoning and context

The production scheme uses one well-mixed thermal zone per storey. Large buildings above the recorded threshold do not receive a core/perimeter subdivision. Party surfaces, context shading and openings are counted per accepted model. A ground storey with no exterior wall can legitimately have zero glazed subsurfaces.

Table 4. Zoning and context

Geometry evidence	Final statistic
Buildings with measured fidelity	Pending
Median / P95 / maximum fidelity	Pending
Refined, coarser or as-drawn preparation	Pending
Storey-snapped buildings and area share	Pending
Large single-zone buildings and energy share	Pending
Party, shading and opening distributions	Pending

7. Use, occupancy and domestic hot water

Ground-storey use is conditional on the final recorded building evidence. Residential ground levels are treated as dwelling space and included in the residential basis. Commercial ground levels follow the verified tertiary regime and are excluded from a residential-only denominator. Mixed or unresolved evidence is not forced into either narrative.

Occupancy reporting separates administrative Padrón population, occupants applied to the model, source/fallback, density cap and plausibility status. Padrón is not live presence. A measured zero, an imputed value and a capped value are different evidence states.

Domestic hot water follows the recorded occupancy and model policy. The cited CTE reference demand of 28 litres/person/day is a 60 °C reference [6]; the model's 50 °C implementation is not claimed to be directly equivalent. Final reporting will state both temperatures and the modelling interpretation.

7.1 Mixed-use energy attribution

Commercial lighting and equipment can be separated by user-defined subcategory. HVAC and DHW are not fully separable by space type under the preserved meter contract. Therefore `residential_site_kwh` is a **residential-attributed upper bound**, not exact residential consumption.

8. HVAC, schedules and climate application

The generated models use the constructions, thermostat schedules, internal-load schedules, space types and HVAC objects bound in the verified template. The final methods table will state the relevant profile values and cite the preserved model/template evidence rather than retyping unsupported assumptions.

Annual weather is read from the recorded Valencia EPW. Design-day objects are selected from the matched DDY. Barometric pressure and site metadata are derived from the active climate contract. The annual Valencia report will not include or annualise the Lecco PALM microclimate slice; that evidence belongs to the separate LECCO_1 event run.

8.1 Parameters to publish

Table 5. Parameters to publish

Parameter family	Publication evidence
Heating/cooling thermostat regimes	Template roles and generated OSM
Infiltration and ventilation assumptions	Verified profile/template evidence
Lighting/equipment schedules	Template roles and subcategory evidence
HVAC system and fuel	Generated model and EnergyPlus tables
DHW demand and temperature	Model preparation record
EPW/DDY site and design days	Climate manifest
Ground and water-mains temperatures	Run climate record

9. Result and denominator contract

Every final metric will be presented with visible name, authoritative field, unit, period, denominator, precision and interpretation limit. Missing optional evidence is an em dash; a measured zero remains zero. Boolean, NaN, infinity and wrong-type values are never formatted as valid measurements.

9.1 Principal energy quantities

Total site energy is the sum of simulated final energy admitted by the result contract. End-use reporting includes space heating, cooling, domestic hot water, fans, pumps, lighting and equipment where recorded. EnergyPlus ABUPS values originate in output meters; user-defined subcategories provide applicable splits [4].

9.2 Area denominators

- **Geometric residential area:** modelled residential-storey basis used by the comparison convention.
- **Cadastral residential area:** Tipo15 administrative area where available.
- **Conditioned area:** all conditioned model area, including applicable non-residential space.
- **Residential-attributed upper bound:** energy remaining after separable commercial categories are removed; not exact attribution.

Whole-site energy divided by residential-only area is a comparison convention. It can include commercial ground-floor energy in the numerator while excluding that floor from the denominator. This is not the physical intensity of all conditioned space.

10. Operational carbon

Operational emissions are derived from simulated final energy using the recorded project factors: 0.331 kgCO₂/kWh_final for electricity and 0.252 kgCO₂/kWh_final for natural gas [5]. The final table will reconcile building-level carbon fields with settled aggregate totals.

The result excludes embodied carbon and is not a measured emissions inventory. Factors may not represent another reporting year, electricity supplier or marginal-emissions question. Carbon figures will remain tied to the annual period and the stated energy basis.

Table 6. Operational carbon

Carbon quantity	Unit	Denominator	Final value
Total operational carbon	tCO ₂ /year	None	Pending
Operational carbon intensity	kgCO ₂ /m ² ·year	Named building area	Pending
HVAC operational carbon	tCO ₂ /year or kgCO ₂ /m ² ·year	Named total/area	Pending

11. Coverage, failures and exclusions

Coverage is reported as successful result count divided by buildings in scope, together with failed, QA-rejected and excluded counts. A high percentage does not prove absence of bias. Missing buildings may differ systematically in size, cluster, geometry or use.

When a common cadastral-area basis exists for represented and missing records, the report may size potential truncation under the recorded convention. Otherwise the correct statement is that truncation is unquantified; it is never “no bias.”

Table 7. Coverage, failures and exclusions

Scope accounting	Final count
Buildings in scope	Pending
Runnable after preflight	Pending
Successful latest records	Pending
Runtime failed	Pending
QA rejected	Pending
Pre-simulation excluded	Pending
Coverage percentage	Pending

Failure and exclusion reasons will be grouped without erasing their references. Repeated attempts are resolved by latest-record semantics; attempts are not double-counted as buildings.

12. Quality assurance and diagnostics

The building QA matrix compares expected model evidence, observed EnergyPlus evidence or accepted method interval, criterion and result. Stored ratio tolerances are reported in reader-facing units: 0.005 as $\pm 0.5\%$, 0.02 as $\pm 2\%$, and annual unmet hours as ≤ 500 h.

Some checks compare the built model with EnergyPlus output. A plausibility-band check evaluates the method against a research range and is not mislabelled as an EnergyPlus cross-check.

12.1 Diagnostic classification

EnergyPlus Severe messages generally require correction [7]. BSEW can accept only a narrowly recorded project-classified benign pattern when all QA checks pass and no Fatal or unexplained Severe remains. Such a building is **ACCEPTED WITH CLASSIFIED DIAGNOSTICS**, not PASS. The final report will disclose message markers, counts and the project-specific acceptance rule.

Table 8. Diagnostic classification

QA/diagnostic evidence	Final result
QA rejected buildings	Pending
Accepted rows with classified benign Severe	Pending
Unexplained Severe	Pending
Fatal	Pending
Occupancy plausibility flags	Pending

13. Aggregate results and cluster interpretation

The final results table will report energy and operational carbon only after settlement. Cluster rows will include building count, residential area, total site energy and the available geometric/cadastral intensities. Rai comparison columns will be shown only where the published comparison exists and the denominator is aligned.

Cluster intensity is energy divided by the corresponding total area. It is not the arithmetic mean of building intensity. A difference from the Rai reference is a method/reference comparison, not calibration against utility observations.

Table 9. Aggregate results and cluster interpretation

Result	Unit and basis	Final value
Total site energy	GWh/year, represented stock	Pending
Space heating	GWh/year	Pending
Cooling	GWh/year	Pending
Domestic hot water	GWh/year	Pending
Whole-site / geometric residential area	kWh/m ² ·year	Pending
Whole-site / cadastral residential area	kWh/m ² ·year	Pending
Whole-site / conditioned area	kWh/m ² ·year	Pending
Operational carbon	tCO ₂ /year	Pending

14. Spatial evidence and exports

The final GeoPackage will be checked against the GeoPackage encoding contract [8] for CRS, feature count, one feature per latest building reference, null-energy representation of failed/excluded records, context fields, derived per-person/per-dwelling fields, cluster/district layers and supplied QGIS style.

The heat map is a publication-oriented image. Its caption will identify run, annual period, metric, denominator, coverage, colour-scale clipping and the simulated-not-measured boundary. Buildings without results will remain visible in a separate status treatment. Distant valid geometries will not be silently cropped from every view.

The Building CSV is the authoritative row-level analysis export after schema correction. Final acceptance will compare its headers and latest-record contents with the grouped User Guide field appendix. Absolute local paths will not appear in the curated publication schema.

14.1 Export integrity

Every DOCX/PDF/HTML report, CSV, GeoPackage, heat map and evidence manifest will receive SHA-256. Signed packages contain an Ed25519 manifest. Hash/signature verification establishes integrity, not scientific validity.

15. Reproducible representative-building selection

The final report will select up to six references by deterministic rules applied to settled latest records. Duplicate selections are removed in the order below and the exact algorithm, source hash and selected references are preserved in the evidence manifest.

1. Successful building nearest the median valid total-site intensity.
2. Successful building nearest the fifth percentile.
3. Successful building nearest the ninety-fifth percentile.
4. Successful building with the greatest total-site energy contribution.
5. Accepted building carrying a recorded occupancy or classified-diagnostic flag, if available.
6. One failed or excluded reference representing the most frequent non-success reason, if available.

For successful examples, the report will show identity, cluster, area bases, energy, carbon, occupancy, geometry, QA status, diagnostic status and links/hashes for readable and technical evidence. The failed/excluded example will not be assigned energy; it will document the failure stage and reason.

Table 10. Reproducible representative-building selection

Selection role	Reference	Evidence status
Median-nearest	Pending settled ledger	Pending
P5-nearest	Pending settled ledger	Pending
P95-nearest	Pending settled ledger	Pending
Highest total contribution	Pending settled ledger	Pending
Flagged accepted example	Pending settled ledger	Pending
Failed/excluded example	Pending settled ledger	Pending

16. Limitations and responsible use

The final interpretation will retain the following limits beside affected findings:

- cadastral geometry, use and area are administrative evidence with measurement/allocation limits;
- one parcel is not always one physical building;
- simplified zoning and integer-storey geometry affect area representation;
- large one-zone-per-storey buildings do not represent core/perimeter diversity;
- difficult/courtyard geometry can be excluded by preparation gates;
- occupancy uses administrative evidence, caps and recorded imputation;
- HVAC and DHW are not exactly separable between residential and commercial spaces;
- operational carbon excludes embodied carbon;
- profile verification is not per-building measurement calibration;
- pilot-building LHS evidence is not a city-total uncertainty band;
- failed/excluded records can create unquantified truncation.

These limitations do not automatically invalidate the study. They define the claims the evidence can support and the additional measurements or sensitivity analyses needed for stronger claims.

17. Provenance, reproducibility and publication package

The final publication package will preserve logical filenames and hashes for the run configuration, process record, ledger, final aggregate, corrected Building CSV, GeoPackage/style, heat map, selected readable reports, EnergyPlus tables, technical evidence and document outputs. Absolute workstation paths will be removed from publication-facing documents and curated exports.

No result will be accepted when duplicate authoritative evidence blocks conflict. A building identity mismatch prevents report generation. A successful-looking row without essential period, result, QA or diagnostic evidence is INCOMPLETE EVIDENCE – DO NOT CITE.

17.1 Evidence manifest

The machine-readable JSON/CSV manifest will contain:

- run, profile, climate, template, policy, stock-source and runner-schema identities;
- final scope/count/period reconciliation;
- source and output SHA-256 values;
- key method parameters and units;
- selected-building algorithm and references;
- figure/table source files;
- software commit and publication date;
- review gates and their PASS/FAIL state.

17.2 Final document control

DOCX and HTML are generated deterministically from this canonical source; PDF is produced from the visually verified DOCX. Every DOCX/PDF page and responsive HTML layout will be inspected, including semantic structure and the WCAG 2.2 accessibility target [9], before final status. The final edition will contain no provisional marker, running total, local path or unsupported claim.

18. References

- [1] Python Software Foundation, “Python 3.13 documentation,” <https://docs.python.org/3.13/>
- [2] NREL, “OpenStudio 3.11.0 release,” <https://github.com/NREL/OpenStudio/releases/tag/v3.11.0>
- [3] EnergyPlus, “EnergyPlus 25.2 documentation,” <https://energyplus.readthedocs.io/en/v25.2.0/>
- [4] U.S. Department of Energy, *EnergyPlus Input Output Reference*, version 25.2, <https://bigladdersoftware.com/epx/docs/25-2/input-output-reference/index.html>
- [5] Instituto para la Diversificación y Ahorro de la Energía, *La bomba de calor en la rehabilitación energética de edificios*, 2023, https://www.idae.es/sites/default/files/documentos/publicaciones_idae/Guias_IDAE_La_Bomba_de_calor_2023_V11.pdf
- [6] Gobierno de España, *Código Técnico de la Edificación, DB-HE*, 2017 edition, https://www.codigotecnico.org/pdf/Documentos/HE/DBHE_201706.pdf
- [7] U.S. Department of Energy, *EnergyPlus Essentials*, version 25.2, <https://energyplus.readthedocs.io/en/v25.2.0/essentials/essentials.html>
- [8] Open Geospatial Consortium, “GeoPackage Encoding Standard,” <https://www.ogc.org/standard/geopackage/>
- [9] W3C, “Web Content Accessibility Guidelines (WCAG) 2.2,” <https://www.w3.org/TR/WCAG22/>